

Nate Cirino

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Graduating May 2027 | Available to start May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Algorithms & Data Structures, Computer Systems, Object-Oriented Design, Foundations of Software Engineering, Digital Design & Computer Architecture, Computer Networks, Systems Security

MACHINE LEARNING PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models in PyTorch on thermal-sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Iterated on model architecture and loss formulation to address severe class imbalance and sensor noise, raising detection performance on held-out evaluation sets
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time latency limits

Sentinel | *Python, FastAPI, LLM tool-use, pgvector, OpenTelemetry, Next.js*

Apr. 2026 – Present

- Built an LLM agent that autonomously triages incidents via multi-step tool-use over Prometheus, Tempo, and Loki telemetry sequences
- Built pgvector embedding retrieval over runbooks and postmortems with prompt caching on frozen schemas; 100% accuracy on an offline eval harness
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree storage engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables, and compaction

EXPERIENCE

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask recommendation backend with a retrieval-augmented pipeline serving personalized workout recommendations to 2K paying users via Bedrock Knowledge Bases
- Evolved the retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, improving recommendation relevance while cutting operational overhead
- Maintained 99.99% uptime across 1–2M monthly model inference calls via AWS WAF, Datadog, and Kubernetes
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows; continue as technical advisor on architecture and agentic direction

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Optimized Expert AI search, retrieval, and filtering across 20+ microservices, improving clinical query relevance at scale
- Integrated Expert AI (GenAI clinical decision support) into CME accrual/redemption microservices, expanding reach to 250K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing LLM workflow
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair physical retail brands: e-labels, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

SKILLS & HONORS

ML & Frameworks: PyTorch, TensorFlow, Pandas, LLM tool-use & prompt caching, RAG / embedding retrieval, pgvector, Vespa, AWS Bedrock, SageMaker, AWS Neuron SDK, FastAPI, Flask

Languages/Tools: Python, C++, C, Rust, Java, TypeScript, JavaScript, Bash, Linux, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, PostgreSQL, GraphQL, React, Next.js, Node.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor